

SRIRAM RAMPELLI

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in **Generative AI, LLMs, and scalable AI applications**. Extensive experience in **LLM fine-tuning, NLP, and multimodal AI**. Published researcher (**IEEE CCWC 2025**) with a keen interest in expanding expertise in **cloud deployment expertise (AWS, GCP, AZURE)**. Proficient in **transformer-based AI models (GPT, BioBERT, Llama)** and **scalable ML pipelines**. Kaggle RSNA 2024 participant, optimizing deep learning for medical imaging.

TECHNICAL SKILLS

- **AI/ML:** PyTorch, TensorFlow, Transformers (Llama, GPT, BioBERT), OpenAI API
- **NLP/LLMs:** spaCy, NLTK, RAG, LlamaIndex, LangChain
- **Data & MLOps:** SQL
- **AI Research:** Model optimization, multimodal AI, clinical NLP

EXPERIENCE

AI Intern – BCG GenAI Job Simulation | Feb 2025 – Feb 2025

- Developed an AI-powered financial chatbot that automated SEC 10-K data extraction, reducing manual effort by 40%.
- Implemented NLP-based structured data extraction using spaCy, pandas.
- Automated financial sentiment analysis increasing decision-making efficiency by **25%**, improving decision-making efficiency for financial teams.

Research Assistant – Lawrence Technological University | May 2024 – Dec 2024

Developed an LLM-powered healthcare chatbot integrating fine-tuned LLaMA & enhanced BioBERT, improving medical Q&A accuracy by 23%.

Designed custom data pipelines to handle noisy clinical text, boosting OCR-based medical condition recognition by 35%.

Published findings at IEEE CCWC 2025, focusing on LLM-driven healthcare automation.

Project Engineer – Wipro Technologies | Nov 2021 – Dec 2022

- Developed an AI-based anomaly detection system, reducing system log errors by 35%.
- Optimized system performance by 27% through ML-driven monitoring framework.
- Managed 300+ test procedures & 100+ Agile sprint projects, ensuring high-quality releases.

AI/ML PROJECTS

RSNA 2024 Lumbar Spine Classification (Kaggle) | [GitHub](#)

- Built CNN-based medical imaging model, achieving 92% accuracy on RSNA dataset.
- Improved model performance through ensemble learning & data augmentation.

AI Chatbot | [GitHub](#)

Leveraged the Natural Language Toolkit (NLTK), a pivotal tool in natural language processing, to develop an advanced chatbot.

This chatbot is designed to deliver detailed and informative responses, showcasing my ability to apply complex NLP techniques to improve user interaction and automate customer service operations.

Chatbot with GPT-2 & NLP | [GitHub](#)

- Developed an AI chatbot leveraging GPT-2 & Wikipedia data, enhancing document Q&A automation.
- Integrated speech-to-text & voice-based interaction for real-time AI responses.

CERTIFICATIONS & PUBLICATIONS

- **Rampelli, S. (2025). *Empowering Healthcare Data Systems with an Innovative Chatbot Application Utilizing Python and Advanced Generative AI Models*. Presented at IEEE CCWC 2025.**
- NVIDIA-Certified Associate: Generative AI & LLMs
- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional

EDUCATION

Master of Science in Computer Science | Lawrence Technological University | Dec 2024